

AI-Powered Smart Energy Consumption Prediction Using Machine Learning for Sustainable Smart Homes

Problem Statement

Modern households often consume more electricity than necessary due to the inability to predict future energy usage. This results in higher electricity bills, energy wastage, and increased environmental impact. The objective of this project is to develop a machine learning regression model that predicts household appliance energy consumption using indoor environmental conditions, weather information, and time-based features. Accurate predictions can support efficient energy management and sustainable smart home solutions.

Dataset Source and Justification

Dataset Name: Appliance Energy Prediction Dataset

Source: Kaggle (<https://www.kaggle.com/datasets/loveall/appliances-energy-predictions>)

This dataset was selected because it contains real-world household energy consumption data along with indoor temperature, humidity, weather conditions, lighting usage, and timestamp information. These features make it well suited for building and evaluating a machine learning model for energy consumption prediction.

Dataset Overview

- **Records:** 19,735
- **Features:** 29
- **Target Variable:** Appliances (Energy Consumption in Wh)
- **Problem Type:** Supervised Machine Learning – Regression
- **Data Characteristics:** Environmental sensor readings, weather data, lighting usage, and timestamp information.

Preprocessing Steps Performed

- Loaded and explored the dataset.
- Examined dataset dimensions, feature names, and data types.
- Generated statistical summaries of numerical features.
- Checked for missing values (none found).
- Checked for duplicate records (none found).
- Detected outliers using the IQR method and retained them because they represent realistic energy consumption patterns.

- Converted the `date` column into numerical time-based features (Year, Month, Day, Hour, Minute, DayOfWeek, and IsWeekend).
- Removed the random features `rv1` and `rv2` as they have no meaningful predictive value.
- Applied **StandardScaler** to standardize numerical features for improved model performance.
- Did not apply class imbalance techniques or SMOTE-Tomek because the project is a regression problem.

Libraries and Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook / VS Code
- Git & GitHub

Challenges Encountered

The dataset contained no missing values or duplicate records, simplifying the preprocessing stage. The main challenge was determining how to handle the `date` feature and outliers. The timestamp was transformed into meaningful time-based features, while outliers were retained because they reflected genuine variations in household energy consumption rather than data errors.